A120601 concise calculus certificate

VERIFIED

Bradley Klee, Mech.An.ika Sol (Open AI)

July 31, 2026

Definition and first terms

$15\,A(x) = 14 + 27\,x + A(x)^6$, with the origin-normalized solution.

Normalization/grammar: `A(x)=1+(3)*T(x); T=x+5*T^2+20*T^3+45*T^4+54*T^5+27*T^6`.

$a(0), \ldots, a(7) = 1, 3, 15, 210, 3510, 65562, 1310901, 27446760.$

Kernel used throughout: $\rho(u) = u\,(1 - 5\,u^1 - 20\,u^2 - 45\,u^3 - 54\,u^4 - 27\,u^5)$, so $x = \rho(T(x))$.

Typogeometry

`1x3` `{1,1}x15` `{1,{1,1}}x75` `{1,1,1}x60`

colored arity geometry; node color distinguishes constructors. Filled endpoints are true leaves; open endpoints are false/empty leaves. For colored grammars, color is genuine constructor data and is not silently converted into a positional zero pattern.

Typogeometric constructor codes and multiplicities: $\Delta_2\colon$ 5, $\Delta_3\colon$ 20, $\Delta_4\colon$ 45, $\Delta_5\colon$ 54, $\Delta_6\colon$ 27.

Coefficient and contour extraction

$$\mathcal{K}_n = \{(k_1, \ldots, k_5) \in \mathbb{Z}^5_{\geq 0} : \sum_{j=1}^{5} j k_j = n - 1\}, \quad K = \sum_{j=1}^{5} k_j,$$

$$a(n) = \frac{3}{n} \sum_{\boldsymbol{k} \in \mathcal{K}_n} \binom{n+K-1}{K} \binom{K}{k_1, \ldots, k_5} 5^{k_1} 20^{k_2} 45^{k_3} 54^{k_4} 27^{k_5}.$$

$$a(n) = \frac{3}{2\pi i n} \oint_\gamma \frac{du}{\rho(u)^n}, \quad n \geq 1.$$

Recurrence

$\sum_{r=0}^{5} P_r(n) a(n+r) = 0$ for $n \geq 1$. The exact degree-5 coefficient polynomials are embedded in `certificate_payload.json` (source: `data/recurrence.json`).

Differential equation

The scalar linear ODE has order 5. It is obtained from the recurrence by the standard Euler-operator substitution. Exact coefficients and boundary polynomial: `data/ode.json`.

Matrices, telescoper certificate, and checks

No compact matrix tuple is shown; see the embedded exact reduction data.

Certificate.

$R(n, u) = N(n, u)/\rho(u)^4$, $\deg_u N = 25$.

Telescoping identity: $\sum_r P_r(n) H_{n+r}(u) = \partial_u (R(n, u) H_n(u))$.

Matrix dimensions: 12×12; remainder 5×6, rank 5, nullity 1. The exact telescoping residual is zero. The recurrence reconstructs all 24 stored terms; 6/6 published OEIS prefix terms match.

Canonical records: `data/tree_model.json`, `data/contour.json`, `data/matrices.json`, `data/recurrence.json`, `data/ode.json`. Embedded payload contains exact sources and SHA-256 matrix identifiers. Compare with the verbose certificate for A120590.